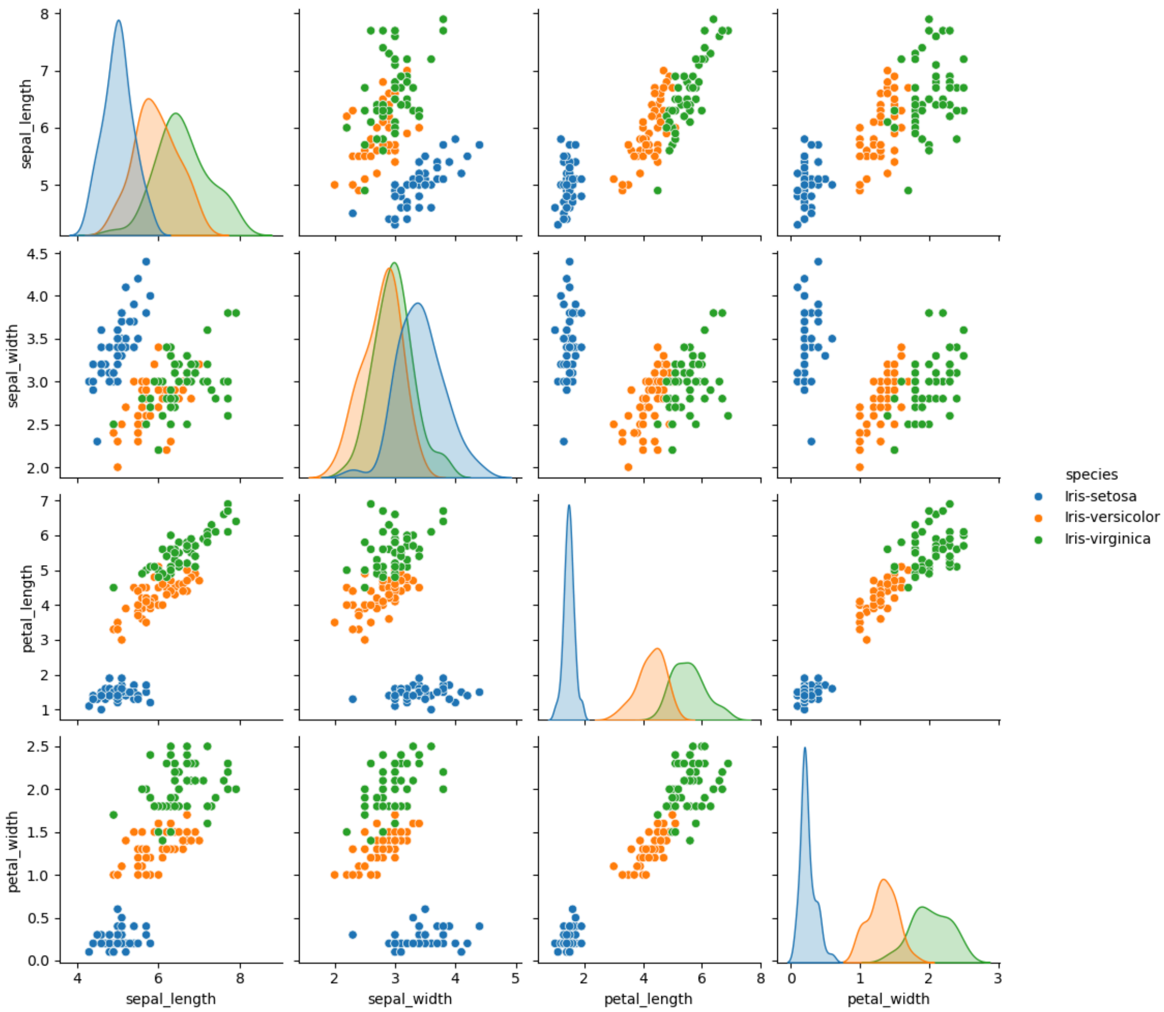
In [1]:
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_mat
```

In [3]:
```
file_path =r"C:\Users\komal\Downloads\Iris.csv - Iris.csv.csv"
data = pd.read_csv( r"C:\Users\komal\Downloads\Iris.csv - Iris.csv.csv")
```

if len(data.columns) > 5: data = data.iloc[:, 1:]

In [15]:
```
data.columns = ['sepal_length', 'sepal_width', 'petal_length', 'petal_width', 's
```

In [17]:
```
sns.pairplot(data, hue='species', diag_kind='kde')
plt.show()
```



In [19]:
```
X = data.iloc[:, :-1]
y = data['species']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_
```

In [21]:
```
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

In [23]:
```
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)
```

Out[23]:
```
▾ RandomForestClassifier
RandomForestClassifier(random_state=42)
```

In [25]:
```
y_pred = model.predict(X_test)
```

In [27]:
```
accuracy = accuracy_score(y_test, y_pred)
print(f'Accuracy: {accuracy:.2f}')
print('\nClassification Report:\n', classification_report(y_test, y_pred))
```

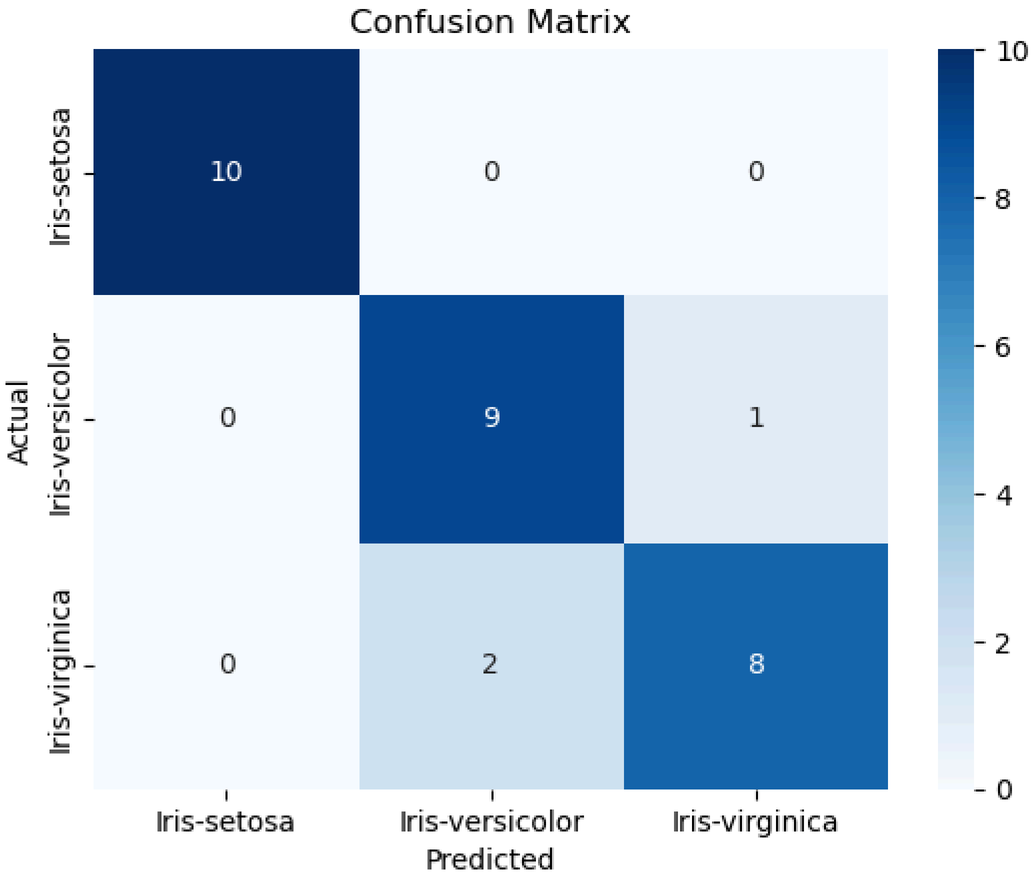
```
Accuracy: 0.90

Classification Report:
                  precision    recall  f1-score   support

    Iris-setosa       1.00      1.00      1.00        10
Iris-versicolor       0.82      0.90      0.86        10
 Iris-virginica       0.89      0.80      0.84        10

       accuracy                           0.90        30
      macro avg       0.90      0.90      0.90        30
   weighted avg       0.90      0.90      0.90        30
```

In [29]:
```
conf_matrix = confusion_matrix(y_test, y_pred)
sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='Blues', xticklabels=np.uniqu
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()
```


Confusion Matrix
Actual
Predicted
Iris-setosa
Iris-versicolor
Iris-virginica
10
0
0
0
9
1
0
2
8
0
2
4
6
8
10

```
In [ ]:
```